CAMERON MICHAEL MATTHEWS

DOCTORAL CANDIDATE- GRADUATE RESEARCH FELLOW

https://matthewscameron.github.io/Cameron-Matthews-Professional-Website/

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LinkedIn:

EXPERTISE

Barley Variety Development

Genomic-Informed Selection Decisions

Multi-Location Trial Execution

Breeding Pipeline Optimization

Breeding Program Management

Cross-Functional Team Direction

Trait Validation & Deployment

EDUCATION

Ph.D. PLANT SCIENCES WITH AN EMPHASIS IN PLANT BREEDING & GENETICS
North Dakota State University
2023 – 2027

BACHELORS OF SCIENCE- BIOCHEMISTRY
Purdue University
2018 – 2022

BACHELOR OF SCIENCE- AGRICULTURAL COMMUNICATION
Purdue University
2018-2022

Objective

Dedicated to advancing regionally adapted crop varieties that deliver high-impact performance for farmers and strengthen agricultural production systems.

Professional Profile

Ph.D. candidate in Plant Breeding & Genetics with experience across public and commercial breeding programs. Graduate Research Fellow leading multi-location trials, advancement decisions, and genomic selection implementation in the NDSU barley breeding program. Nationally recognized leader (NAPB Borlaug Scholar; ASA, CSSA, SSSA Future Leaders in Science Award, NAPB GSWG Chair).

PROFESSIONAL EXPERIENCE

NORTH DAKOTA STATE UNIVERSITY — DEPARTMENT OF PLANT SCIENCES FARGO, ND

Graduate Research Fellow | May 2024 – Present

- Directed nine annual multi-location field trials, coordinating planting, phenotyping, harvest, and data integrity across sites
- Implemented disease nursery harvest protocol, reducing labor time by 50%
- Made final selection decisions on 1,200 early-generation lines per cycle from pre-selected candidates
- Supervised and trained 4+ field and lab staff; led hiring decisions and assigned weekly workloads to maintain operational timelines

Graduate Research Assistant | Jan. 2023 – Present

- Developed and delivered genomic selection models for kernel discoloration resistance, grain protein content, and dormancy length, advancing implementation in the NDSU Barley Breeding Program
- Conducted a preliminary evaluation of arabinoxylan variability, assessing its potential as a selection target for future breeding efforts

PURDUE UNIVERSITY — SCARF UNDERGRADUATE RESEARCH FELLOW (STRIGA RESISTANCE PROGRAM) WEST LAFAYETTE, IN | JAN. 2022 – DEC. 2022

- Screened sorghum inbred lines under controlled conditions to identify resistance mechanisms and support advancement decisions
- Delivered resistance screening results across two resistance timepoints, influencing subsequent gene identification efforts.

CORTEVA AGRISCIENCE — INTEGRATED FIELD SCIENCE INTERN JOHNSTON, IA | MAY 2021- DEC. 2021

- Conducted transgene efficacy evaluations in corn and soybean trials.
- Led field teams of 4-8 personnel in insect, disease, and agronomic data collection.

CORTEVA AGRISCIENCE — FIELD RESEARCH INTERN WINDFALL, IN | MAY 2020- AUG. 2020

- Supported commercial hybrid wheat production; evaluated male sterility systems to inform hybrid development decisions
- Led agronomic data collection and harvest within commercial breeding workflows, ensuring data accuracy and efficiency

AWARDS

2026 CSSA Gerald O. Mott Award Recipient

2026 NDSU Prem Jauhar Crop Science Research Award Recipient

2026 NDSU AgTech Hackathon Plant Breeding Track 2ND Place Team

2025 ASA, CSSA, & SSSA Future Leaders in Science Award Recipient

2025 & 2024 North Dakota State University Frank Bain Scholarship Recipient

2022 National Association for Plant Breeding Borlaug Scholar

ASSOCIATIONS

2023-Present ASA, CSSA, SSSA Member

2022-Present National Association for Plant Breeding Member

COURSES TAUGHT

Teaching Assistant for Genetics Laboratory (BIOL/PLSC 315L), supporting laboratory instruction in foundational genetic principles and applied experimental techniques.

Teaching Assistant for World Food Crops (PLSC 110), facilitating instruction on crop growth, global production systems, management practices, and agricultural processing.

LEADERSHIP

- 2026 41st Annual PSGSS Symposium Organizing Committee Member
- 2026-2027 National Association for Plant Breeding Graduate Student Working Group Chair
- 2025–2026 National Association for Plant Breeding Graduate Student Working Group Vice Chair
- 2024–2025 National Association for Plant Breeding Graduate Student Working Group Secretary
- 2023–Present National Association for Plant Breeding Borlaug Scholarship Committee Member

PROFESSIONAL DEVELOPMENT

- 2025 Bayer 4 You Mentee
- 2019–2020 Purdue Issues 360 Class Seven Fellow

COLLABORATIONS

- Co-developed a 3D-printed seed counter (Dr. Paulo Flores Lab, NDSU ABEN) for small seed lots (≤100 seeds), reducing counting time by 25% and error rates by 15%

AGRICULTURAL MEDIA PUBLICATIONS

- Matthews, C. (2022, December 6). Why future crops could look different. Farm Progress; Indiana Prairie Farmer. https://www.farmprogress.com/crops/why-future-crops-could-look-different

SCIENTIFIC PRESENTATIONS/PUBLICATIONS

- 2026 Plant and Animal Genome 33- Genomic Selection Decouples Unfavorable Linkage Between Kernel Discoloration Resistance and Low Grain Protein Content in Malting Barley
- 2026 American Malting Barley Association (AMBA) Barley Improvement Conference – From Sugars to Suds: Genotypic Variation in Arabinoxylan Content and Structure
- 2025 National Association for Plant Breeding- Cracking the Code: Dissecting Kernel Discoloration and Grain Protein Content Using GWAS and Genomic Selection in Barley
- 2025 Plant and Animal Genome 32- Unraveling QTLs for Kernel Discoloration Resistance in Malting Barley
- 2024 14th International Barley Genetics Symposium Oral Presentation- What QTLs are affecting Kernel Discoloration Resistance and Grain Protein Content in Spring Malting Barley (Hordeum vulgare L.)?
- 2024 National Association for Plant Breeding Conference Poster Presentation- Identification of QTLs Impacting Kernel Discoloration Resistance and Grain Protein Content in Barley (Hordeum vulgare L.)
- 2022 Purdue Office of Undergraduate Research Summer Research Symposium Poster Presentation- Laboratory screening of sorghum lines for incompatibility, a post-attachment resistance mechanism to the parasitic weed *Striga hermonthica*
- College of Agriculture 2022 Purdue Office of Undergraduate Research Fall Research Expo Research Talk: Screening of Sorghum Lines for Incompatibility, a Post-attachment Resistance Mechanism to the Parasitic Weed *Striga hermonthica*